

Project Proposal Report

Automated MCQ Generator from Textbook Using Natural Language Processing (NLP)

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Project Title

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Introduction:

Multiple-choice questions (MCQs) are widely used in educational assessment because they allow instructors to evaluate students efficiently across a broad range of topics. However, manually preparing high-quality MCQs from textbooks is time-consuming and requires considerable effort. An instructor needs to identify important concepts, formulate meaningful questions, and create plausible but incorrect distractors while ensuring that the questions remain relevant to the source material.

Recent advances in Natural Language Processing (NLP) and Transformer-based language models have made automatic question generation possible. Existing research has explored approaches based on models such as T5, BERT, BART, and FLAN-T5 for generating questions and distractors.

Our proposed project aims to develop an NLP-based system that can automatically generate MCQs from textbook text or PDF documents. The system will first extract and preprocess the textual content, identify important concepts or answer candidates, generate questions using a Transformer-based question-generation model, and then generate plausible distractors. The final output will consist of a complete MCQ containing a question, one correct answer, and three incorrect but plausible options.

The project will particularly focus on textbook-oriented question generation, where the generated questions should remain faithful to the provided textbook content. This is important because a generated question may be grammatically correct but still be irrelevant to the source material or contain an incorrect answer.

Probable Methodology:

The proposed system will consist of the following major stages:

1.Textbook/PDF Input:

The user will provide either:

- a textbook chapter in PDF format, or
- raw textbook text.

For PDF input, the system will extract the textual content using a PDF text-extraction library. Unnecessary elements such as page numbers, repeated headers, symbols, and formatting artifacts will be removed during preprocessing.

2.Text Preprocessing:

The extracted text will be cleaned and prepared for NLP processing.

The preprocessing steps will include:

1. Removing unnecessary symbols and formatting.
2. Sentence segmentation.
3. Tokenization.
4. Removal of irrelevant textual elements.
5. Chunking long passages into smaller sections.

Chunking is necessary because Transformer models have input-length limitations. The reference paper similarly divides long input into smaller chunks before performing keyword extraction and question generation.

3.Important Concept / Answer Extraction:

The system will identify important concepts from each textbook passage. The extracted concepts will be treated as candidate answers around which questions can be generated.

4.Question Generation:

A Transformer-based sequence-to-sequence model will generate questions from the textbook passage and selected answer.

A possible formulation is:

Input: Context + Answer

Output: Question

SQuAD 2.0 can be used for fine-tuning or adapting the question-generation model because it provides context-question-answer relationships. The Hugging Face dataset currently contains 130,319 training examples and 11,873 validation examples.

5. Distractor Generation:

After generating the question and identifying the correct answer, the system will generate three plausible incorrect answers.

The generated distractors must satisfy two conditions:

1. They must be incorrect according to the textbook context.
2. They must be semantically plausible enough to function as real distractors.

This is one of the most challenging components of automatic MCQ generation. Research has shown that distractors need to be both incorrect and sufficiently plausible.

6.MCQ Validation:

Before presenting the final MCQ, the system will perform several validation checks:

- Question is grammatically valid.
- Correct answer occurs in or is supported by the source text.
- Distractors are different from the correct answer.
- Distractors are not duplicated.
- Distractors are sufficiently related to the question.
- The question is relevant to the textbook passage.
- The generated question does not contain unsupported information.

Invalid questions will be removed or regenerated.

7.Final MCQ Generation:

The final system will present MCQs in the following format:

For example,

Question:

Which protocol provides reliable data delivery?

- A. UDP
- B. TCP
- C. IP
- D. ARP

Correct Answer: B. TCP

The system may allow the user to specify the desired number of questions.

Expected Output:

The expected output of the proposed system is a set of automatically generated MCQs from a user-provided textbook or PDF.

For each question, the project is expected to generate:

- A grammatically correct question.
- One correct answer.
- Three plausible distractors.
- The source context from which the question was generated.
- The correct answer identified separately.

We expect to generate multiple questions covering different important concepts rather than repeatedly generating questions about the same keyword through this project.

The final system may also provide a downloadable or printable MCQ set for students and instructors.

Discussion:

The proposed system aims to reduce the time and effort required to prepare textbook-based assessments. Instead of manually reading an entire chapter and preparing every question, an instructor will be able to upload the textbook chapter and automatically obtain a collection of MCQs.

The main technical challenge will be maintaining the relationship between the generated MCQ and the original textbook content. A question can be grammatically correct but still be factually incorrect or unsupported by the source. Therefore, the proposed system will use source-context validation before presenting the final question.

Another major challenge is distractor generation. Generating an incorrect option is relatively easy, but generating an incorrect option that appears reasonable to a student is considerably more difficult. Existing research identifies distractor generation as a major challenge in automatic MCQ generation.

The proposed project will therefore combine Transformer-based question generation with semantic-based distractor generation and post-generation validation.

References:

- [1] Neupane, A., Chaudhari, S., & Shah, S. (2025): An Automated MCQ Generator using NLP.
- [2] Chidinma A. Nwafor & Ikechukwu E. Onyenwe: An Automated Multiple-Choice Question Generation Using Natural Language Processing Techniques
- [3] Dr. Rajesh Satri, Gorige Anusha, & C. Sony (2025): AI-Driven MCQ Generation Using NLP
- [4] Pranav Rajpurkar, Robin Jia, & Percy Liang (2018): Know What You Don't Know: Unanswerable Questions for SQuAD
- [5] SQuAD 2.0
- [6] <https://rajpurkar.github.io/SQuAD-explorer/>